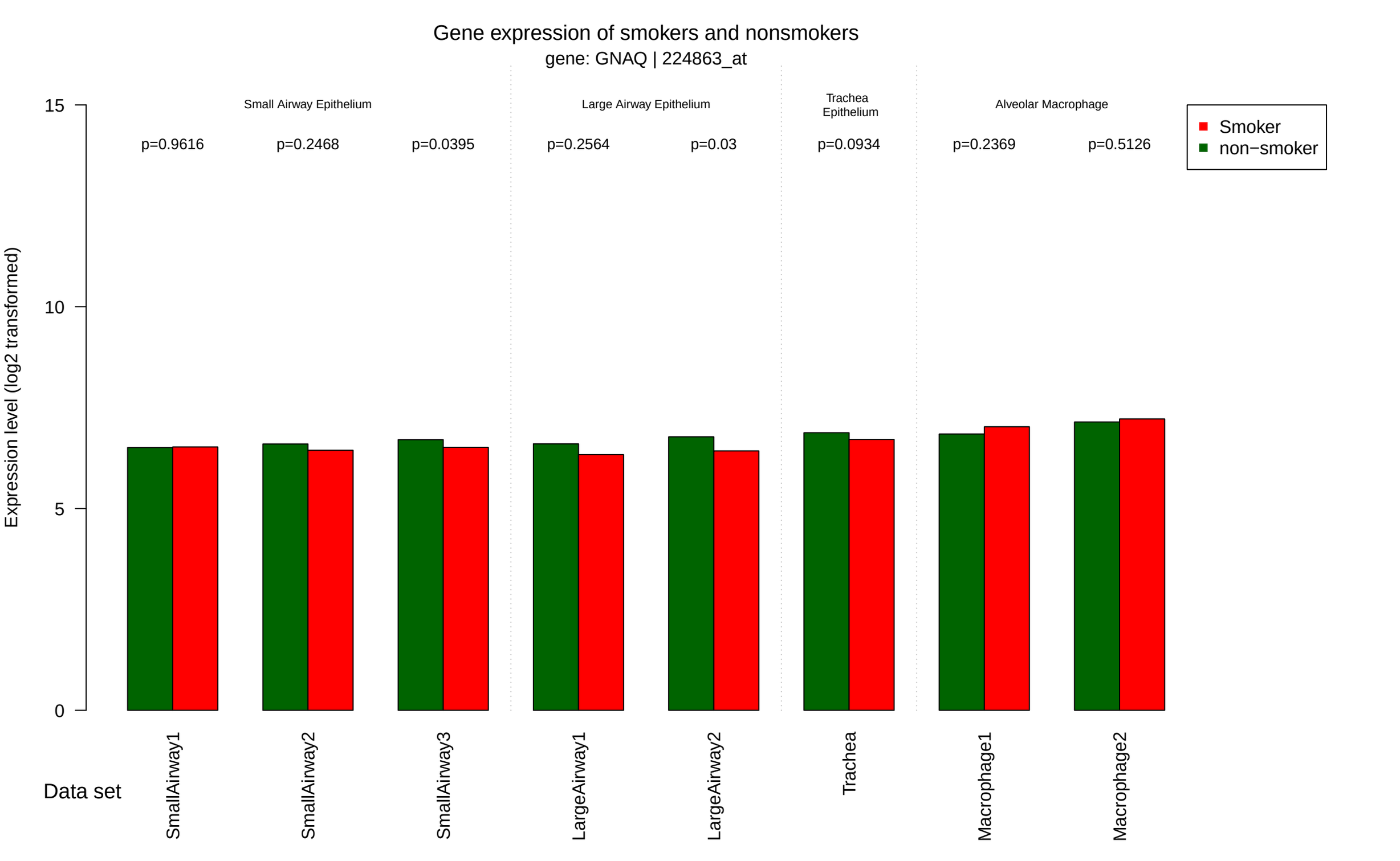
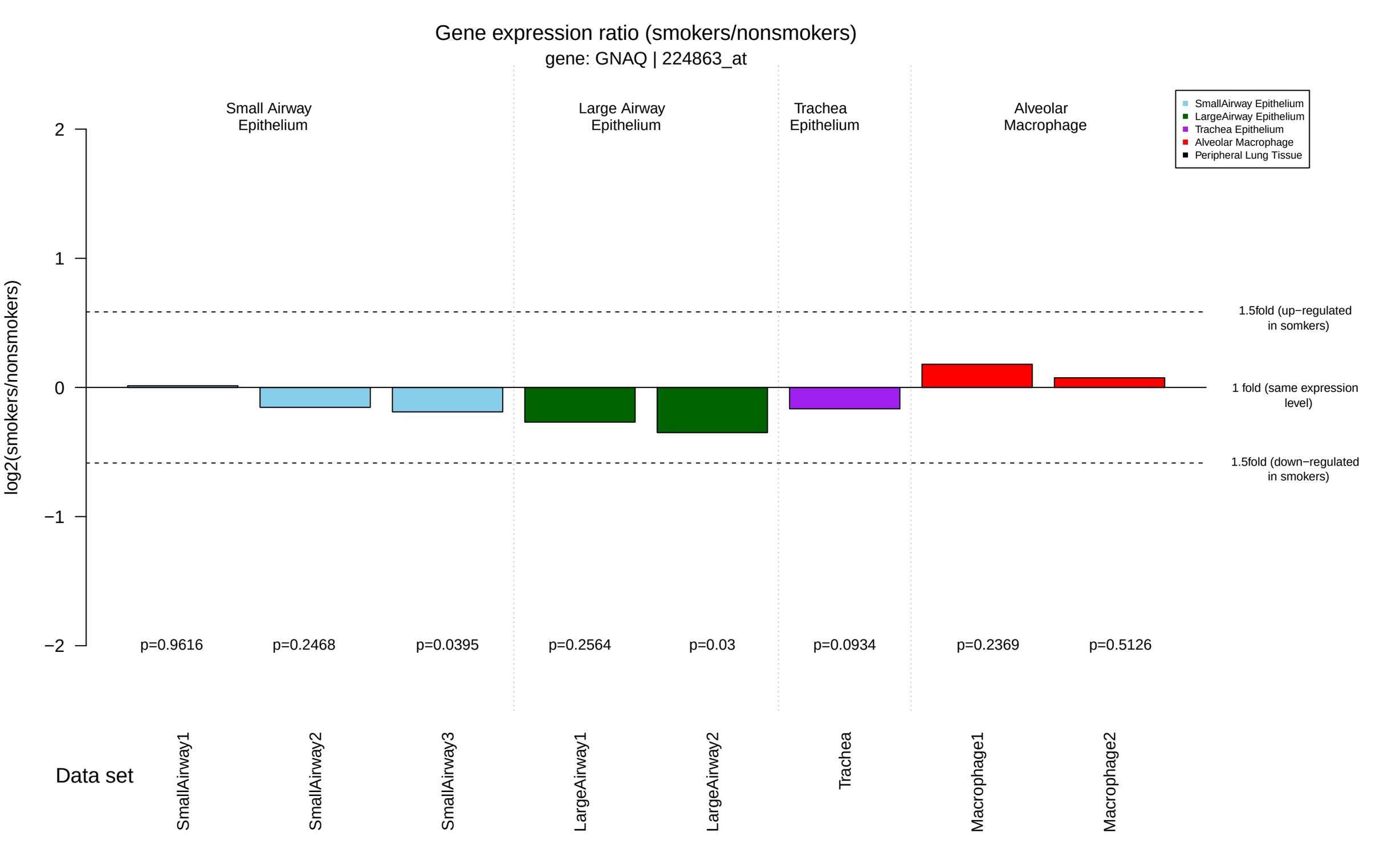
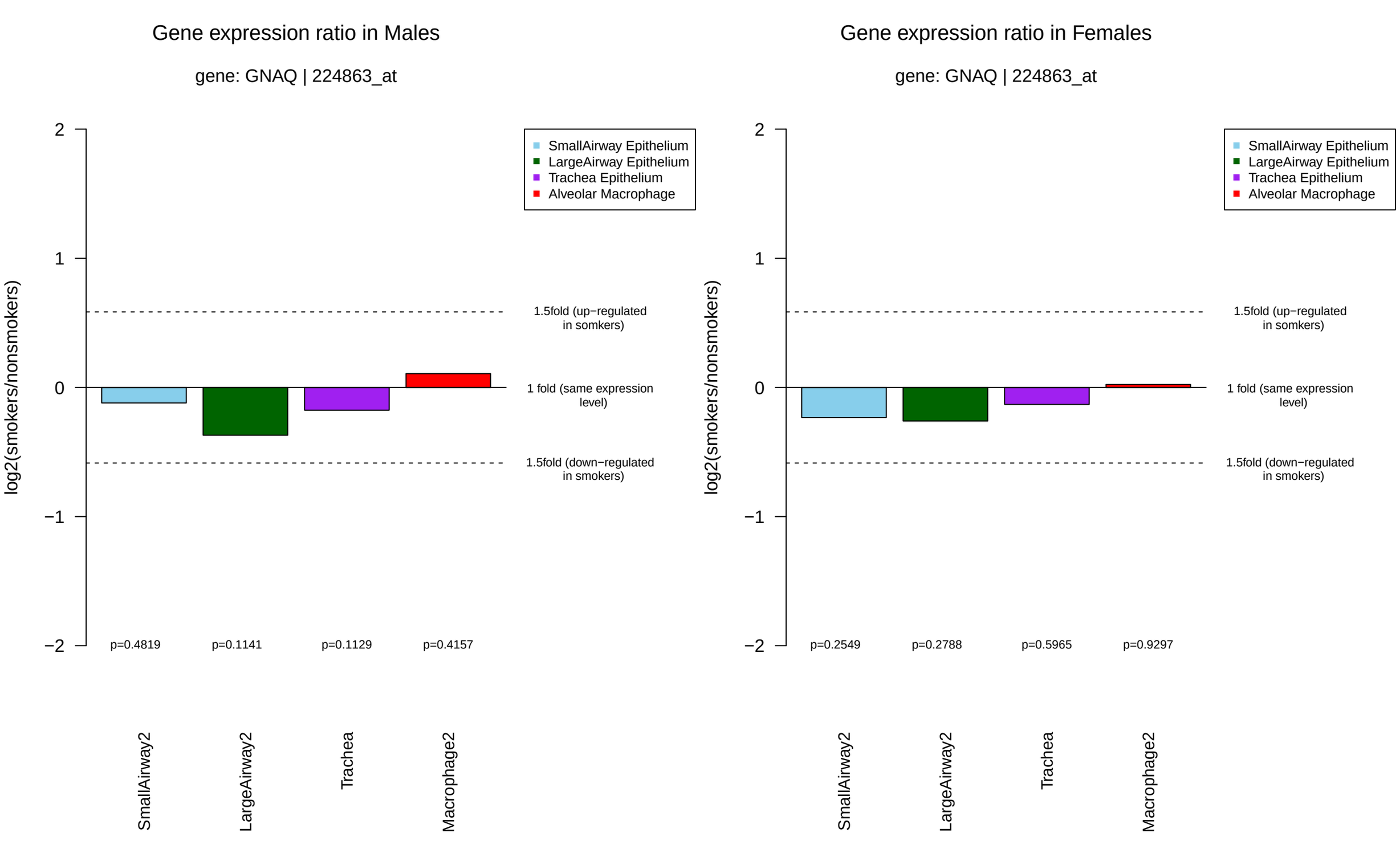
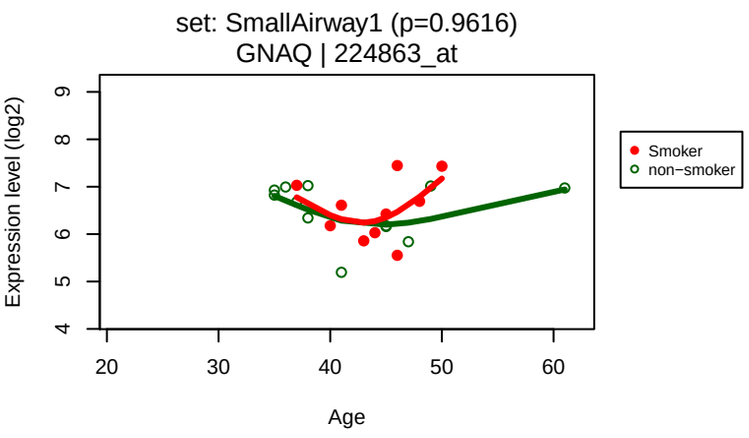
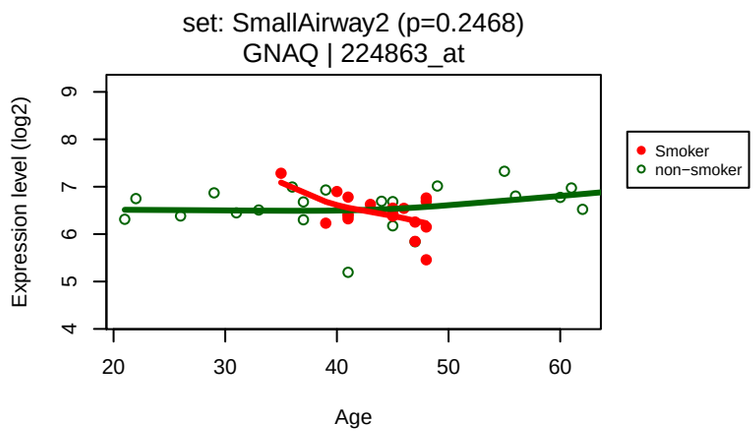
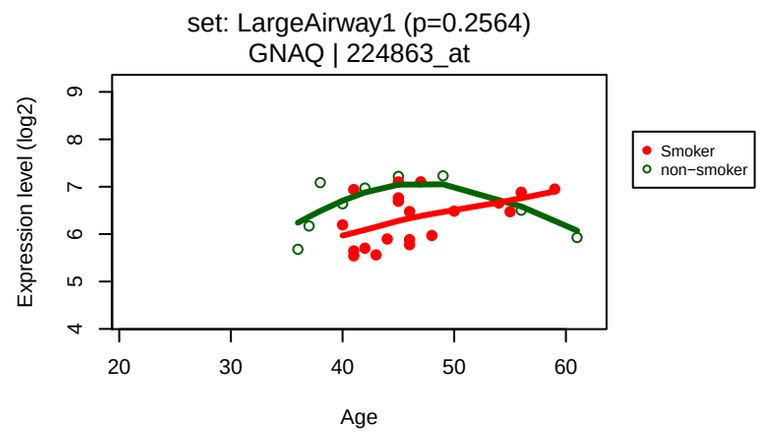

Gene expression of smokers and nonsmokers
gene: GNAQ | 224863_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9616
p=0.2468
p=0.0395
p=0.2564
p=0.03
p=0.0934
p=0.2369
p=0.5126
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

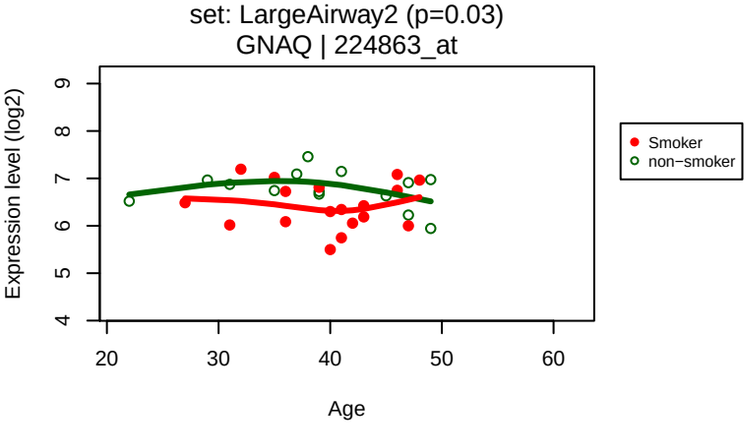

Gene expression ratio (smokers/nonsmokers)
gene: GNAQ | 224863_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9616
p=0.2468
p=0.0395
p=0.2564
p=0.03
p=0.0934
p=0.2369
p=0.5126
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

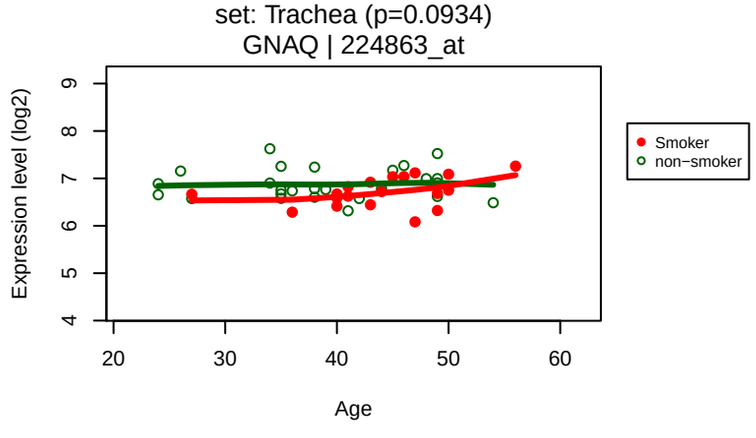

Gene expression ratio in Males
gene: GNAQ | 224863_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4819
p=0.1141
p=0.1129
p=0.4157
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAQ | 224863_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2549
p=0.2788
p=0.5965
p=0.9297
SmallAirway2
LargeAirway2
Trachea
Macrophage2

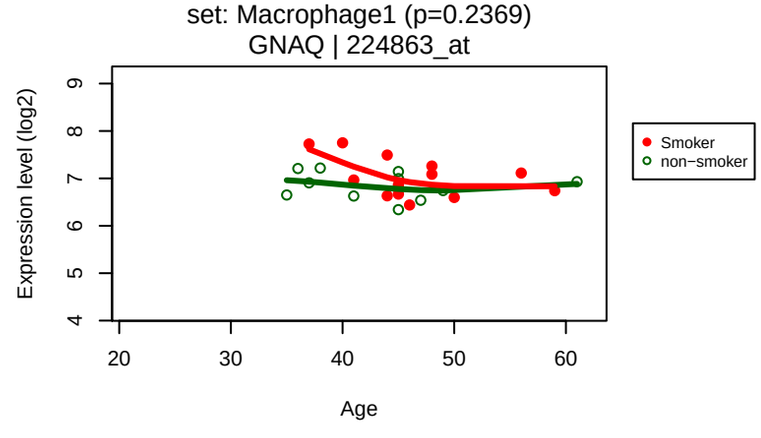

set: SmallAirway1 (p=0.9616)
GNAQ | 224863_at
Expression level (log2)
Age
Smoker
non−smoker

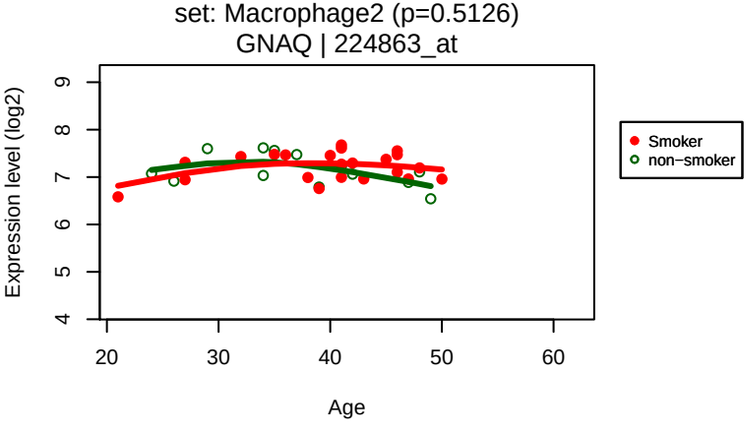

set: SmallAirway2 (p=0.2468)
GNAQ | 224863_at
Expression level (log2)
Age
Smoker
non−smoker

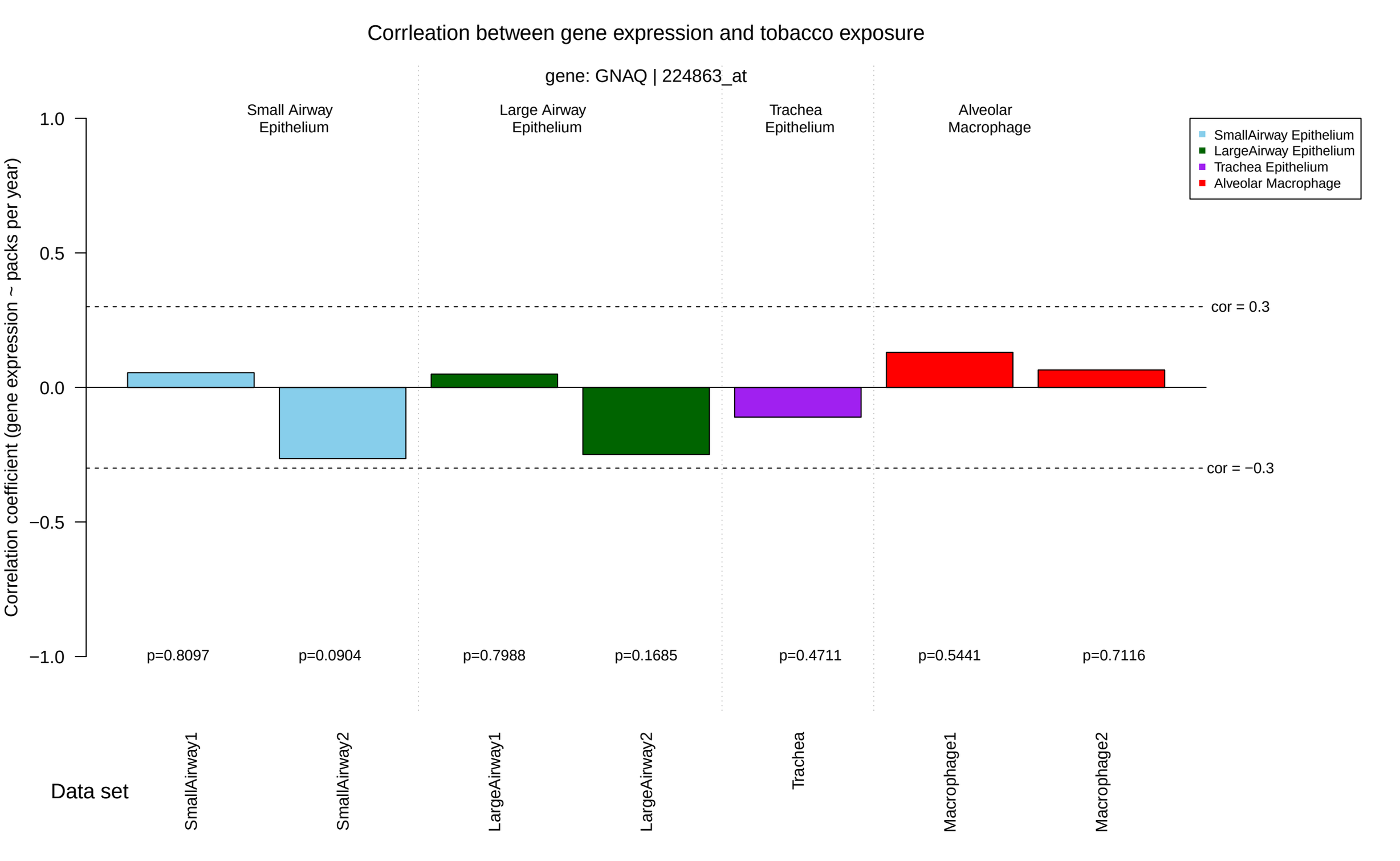

set: LargeAirway1 (p=0.2564)
GNAQ | 224863_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.03)
GNAQ | 224863_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0934)
GNAQ | 224863_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2369)
GNAQ | 224863_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5126)
GNAQ | 224863_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAQ | 224863_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8097
p=0.0904
p=0.7988
p=0.1685
p=0.4711
p=0.5441
p=0.7116
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2